MISH 33
Floor Openings, Open-Sided Surfaces, and Ramps

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Open Floors & Holes

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MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

FLOOR OPENINGS, OPEN-SIDED SURFACES, AND RAMPS

Number

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1.0 PURPOSE

To define protective measures to be implemented on construction sites that involve temporary conditions of floor openings and open sided work surfaces where there exists danger of employees or materials falling into or off of working surfaces. Further, to ensure compliance with Occupational Safety and Health Administration (OSHA) and Washington Industrial Safety and Health Act (WISHA).

2.0 RESPONSIBILITY

2.1 The Key Supervisor shall be responsible for ensuring compliance with this procedure.

2.2 Employees shall be responsible for compliance with the provisions of this procedure.

3.0 DEFINITIONS

Floor Hole: Ann opening measuring less than 12 inches, but more than one inch in its least dimension in any floor, roof, or platform through which materials may fall.

Floor Opening: An opening measuring 12 inches or more in its least dimension in any floor, roof, or platform through which persons may fall.

Runaway: A passageway for persons, elevated above the surrounding floor or ground level.

Standard Railing: A railing that consists of a top rail, intermediate rail, toe board, and posts, and shall have a vertical height of 42 inches from upper surface top rail to floor, platform, runway, or ramp. I shall be able to withstand at least the minimum of 200 pounds top rail pressure with a minimum of deflection.

Toeboard: A vertical barrier at floor level erected along exposed edges of a floor opening, wall opening, platform, runway, or ramp to prevent material falling.

Unprotected Side/Edge: Any side or edge where there is not a wall or handrails.

4.0 GENERAL REQUIREMENTS

4.1 Floor openings, open sided surfaces and ramps shall comply with the following statutory requirements:

• 29 Code of Federal Regulations (CFR) 1926, Subpart M; Floor and Wall Openings

• Washington Administrative Code (WAC) 296-155, Part K; Floor Openings, Wall Openings and Stairways

4.2 FLOOR OPENINGS

4.2.1 Floor openings shall be guarded by a standard railing and toe boards on all exposed sides except at entrances to stairways.

4.2.2 Hatchways and chute openings shall be guarded by one of the following:

• Hinged covers of standard strength and construction and a standard railing with only one exposed side.

• A removable standard railing with toe board on no more than two sides of the opening and fixed standard railings with toe boards on all other exposed sides. The removable railing shall be kept in place when eth opening is not in use and shall be hinged or mounted to be conveniently replaceable.

4.2.3 Pits and trap door floor openings shall be guarded by floor opening covers of standard strength and construction. While the cover is not in place, the pit or trap openings shall be protected on all exposed sides by removable standard railings.

4.2.4 Manhole floor openings shall be guarded by standard covers which need not be hinged in place. While the cover is not in place, the manhole opening shall be protected by standard railings.

4.2.5 All floor opening covers shall be capable of supporting the maximum potential load but never less than two hundred pounds (with a safety factor of four).

4.2.5.1 The cover shall be recessed to conform to the level of the surrounding floor or to be flush with the perimeter of the opening.

4.2.5.2 The cover shall be secured by fastening devices to prevent unintentional removal.

4.2.5.3 If it becomes necessary to remove the cover, a monitor shall remain at the opening until the cover is replaced. The monitor shall advise persons entering the area of the hazard, shall prevent exposure to the fall hazard and shall perform no other duties.

4.2.5.4 Floor opening covers shall be CLEARLY marked, “HOLE” OR “FLOOR OPENING COVER.”

4.2.6 Floor holes into which persons can accidentally walk, shall be guarded by either a standard railing with standard toe boards on all exposed sides, or a floor hole cover of standard strength and construction that is secured against accidental displacement. While the cover is not in place the floor hole shall be protected by a standard railing.

4.3 OPEN SIDED SURFACES

4.3.1 Every open sided floor or platform or surface 4 ft. or more above adjacent floor or ground level shall be guarded by a standard railing on all open sides except where there is an entrance to a ramp, stairway, or fixed ladder. The railing shall be provided with a standard toe-board beneath the open sides wherever persons can pass, there is moving machinery, or there is equipment with which falling materials could create a hazard.

4.3.2 Runways shall be guarded by standard railing on all open sides 4 ft. or more above floor or ground level. Wherever tools, machine parts, or materials are likely to be used on the runway, a toe-board shall be provided on each exposed side.

4.3.3 Where employees entering runways become exposed to machinery, electrical equipment or other dangers that are not a falling hazard, additional guarding shall be provided.

4.3.4 Regardless of height, open sided surfaces, walkways, platforms, runways above or adjacent to dangerous equipment shall be guarded with a standard railing and toe-board.

4.4 RAMPS

4.4.1 Ramps (runways) shall be a minimum of eighteen inches in width.

4.4.2 Ramps shall be secured so that they cannot be accidentally displaced.

4.4.3 Ramps shall be provided with standard railing when located four feet above the ground or floor level.

4.4.4 Ramp specifications: Ramps (runways) and walkways shall not be inclined more than twenty degrees from horizontal and when inclined shall be cleated or otherwise treated to prevent a slipping hazard on the walking surface.

4.5 FALL PROTECTION: Where guardrails cannot be provided and the potential for fall injuries exist, employees shall be protected in accordance with MISH 11.

5.0 PROCEDURE

Not Applicable. .

6.0 RECORDS

Not Applicable

7.0 REFERENCES

Not Applicable

8.0 ATTACHMENTS

Not Applicable

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